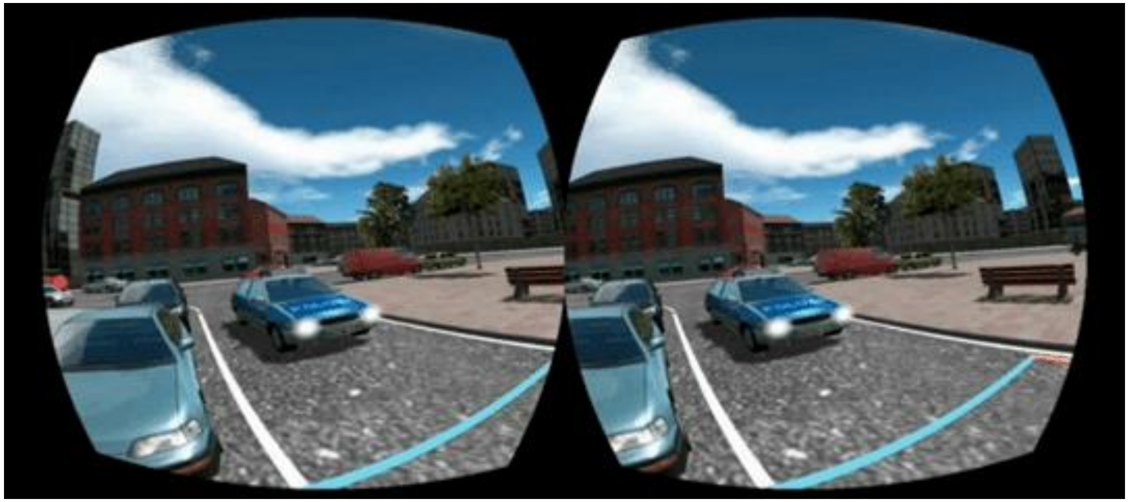
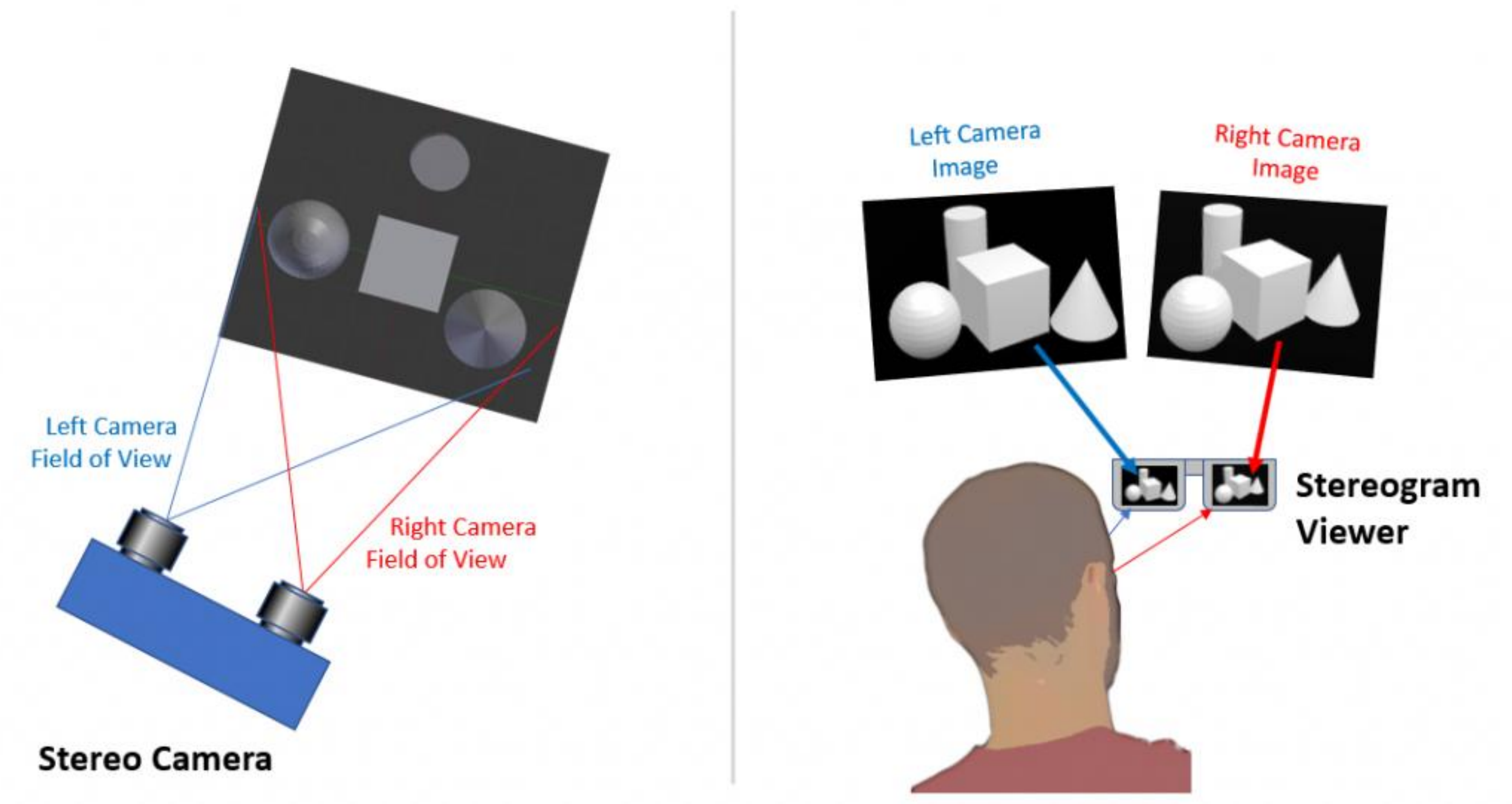
Stereo Graphics

Stereoskopik Görüntü

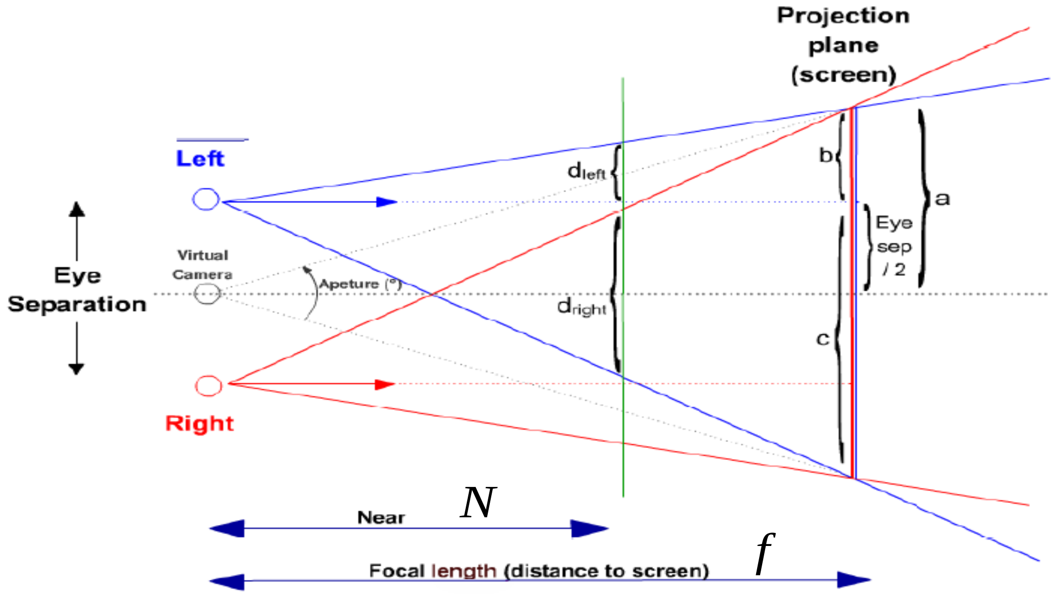


Sol göz Sağ göz

Stereoskopik Görüntü

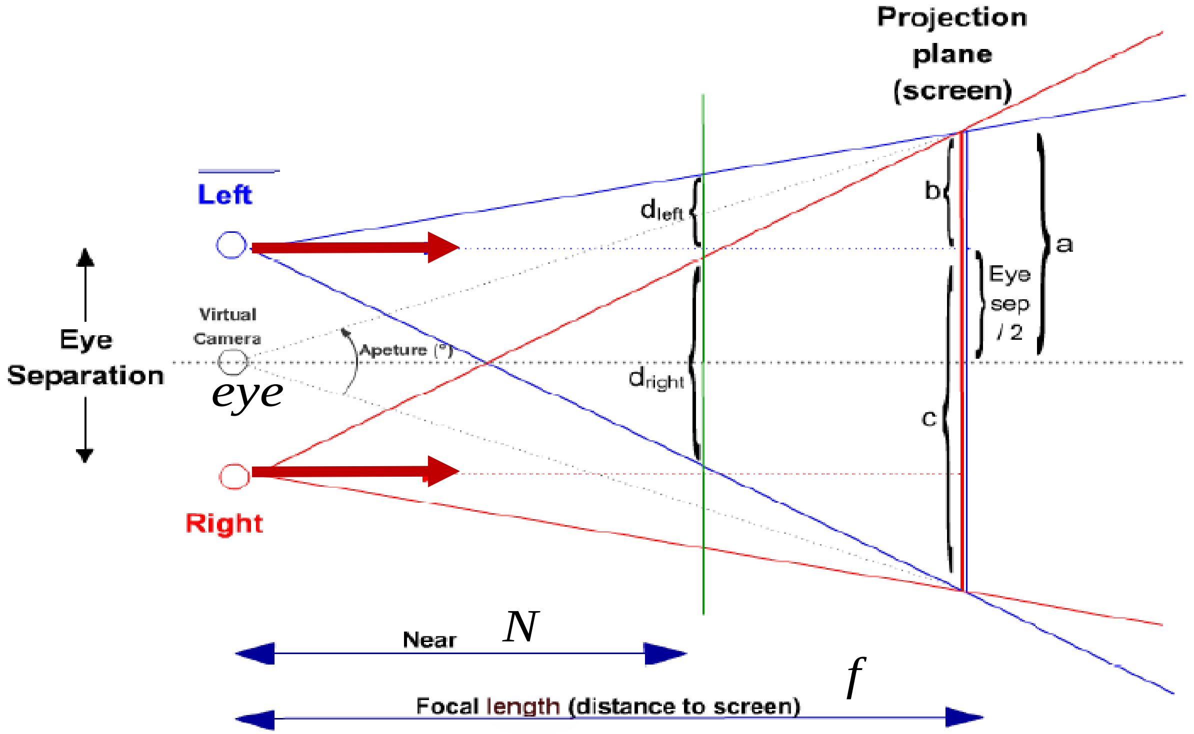


Projection



Focal length (f) is the distance of the camera to the objects that the user wants to focus.

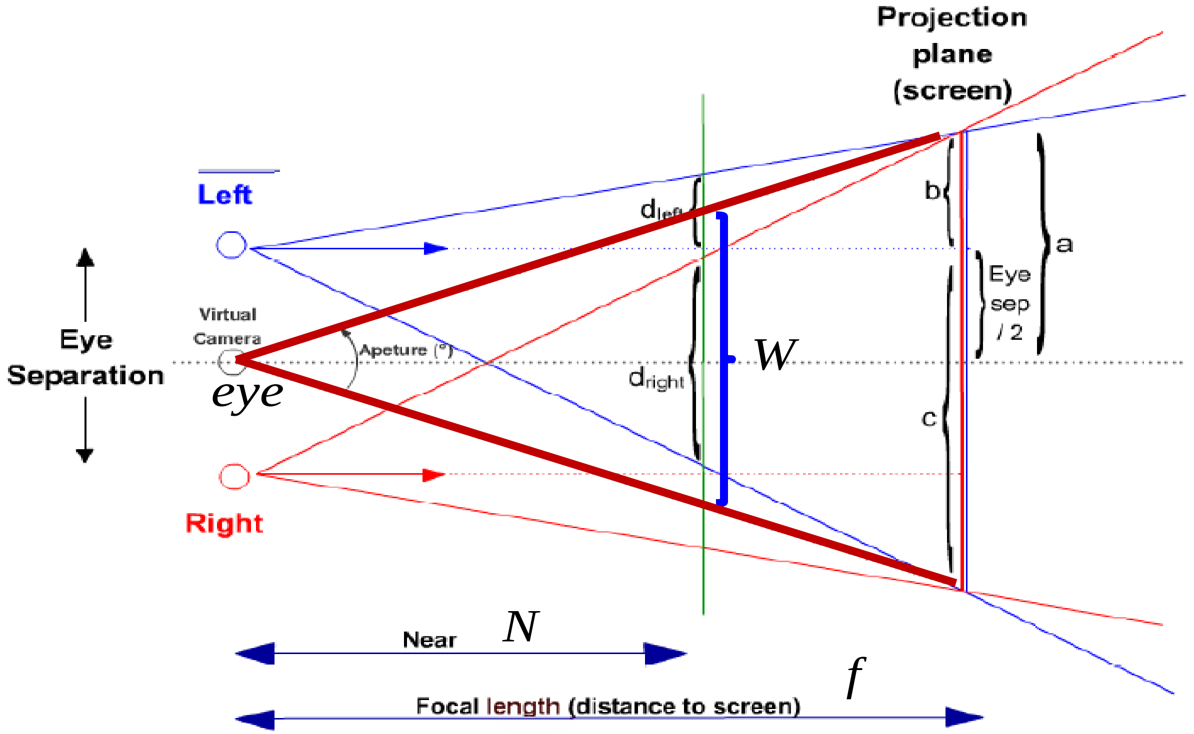
f is between near plane (N) and far plane (F).



Eye Seperation (es) is the distance between the two cameras.

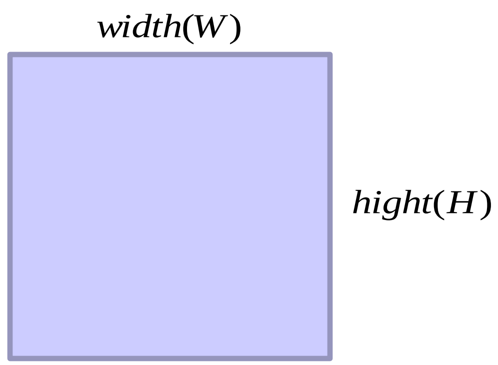
Virtual camera (eye) is in the middle of the two cameras.

Lookat directions of the cameras are parallel to each other.

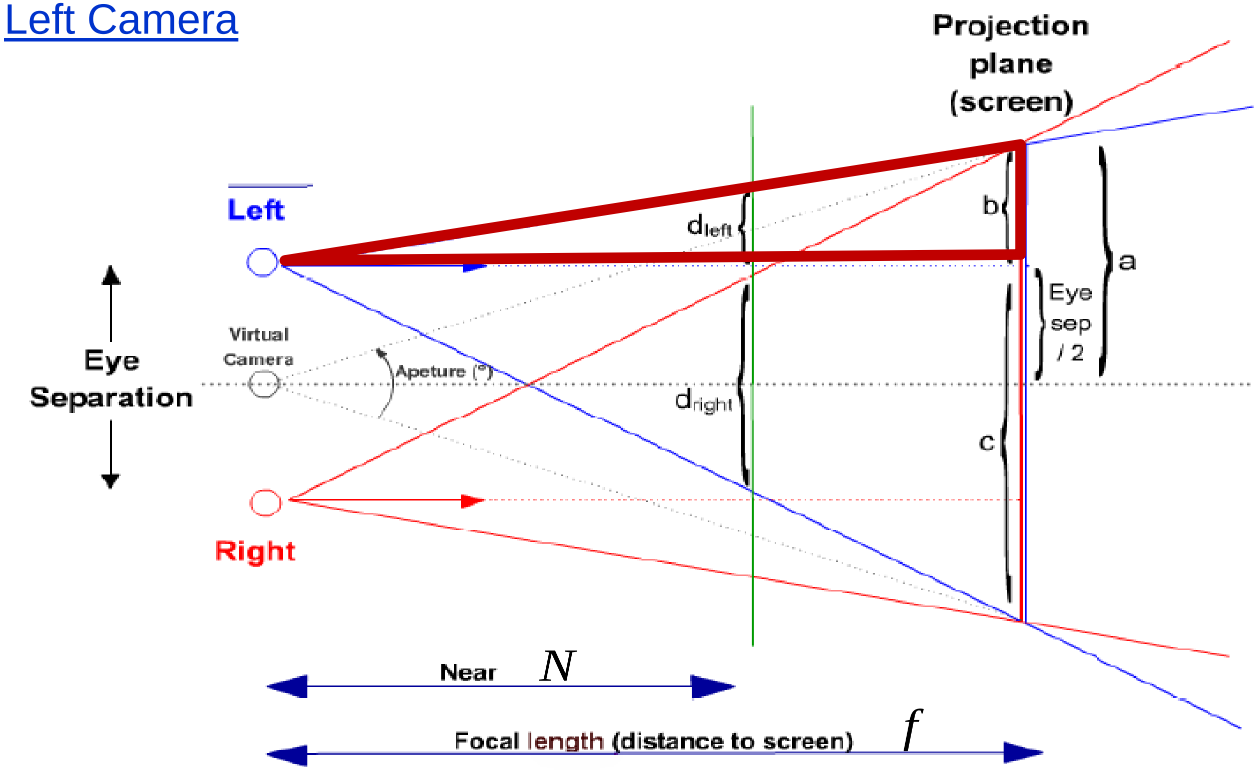


aspect ratio = $\dfrac{W}{H}$

You may assume that aspect ratio is equal to 1.



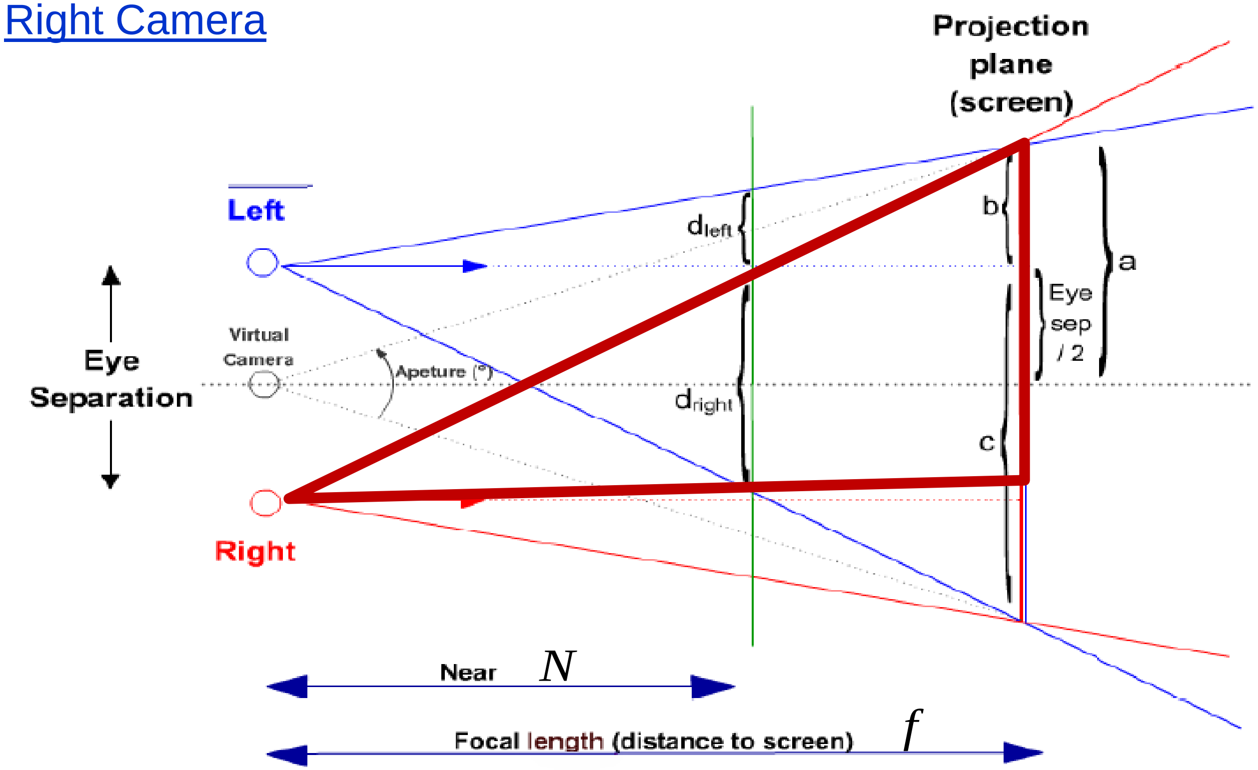
Left Camera



$$\frac{d_{left}}{b} = \frac{N}{f}$$

$$L = -d_{left} = -b\frac{N}{f}$$

Right Camera



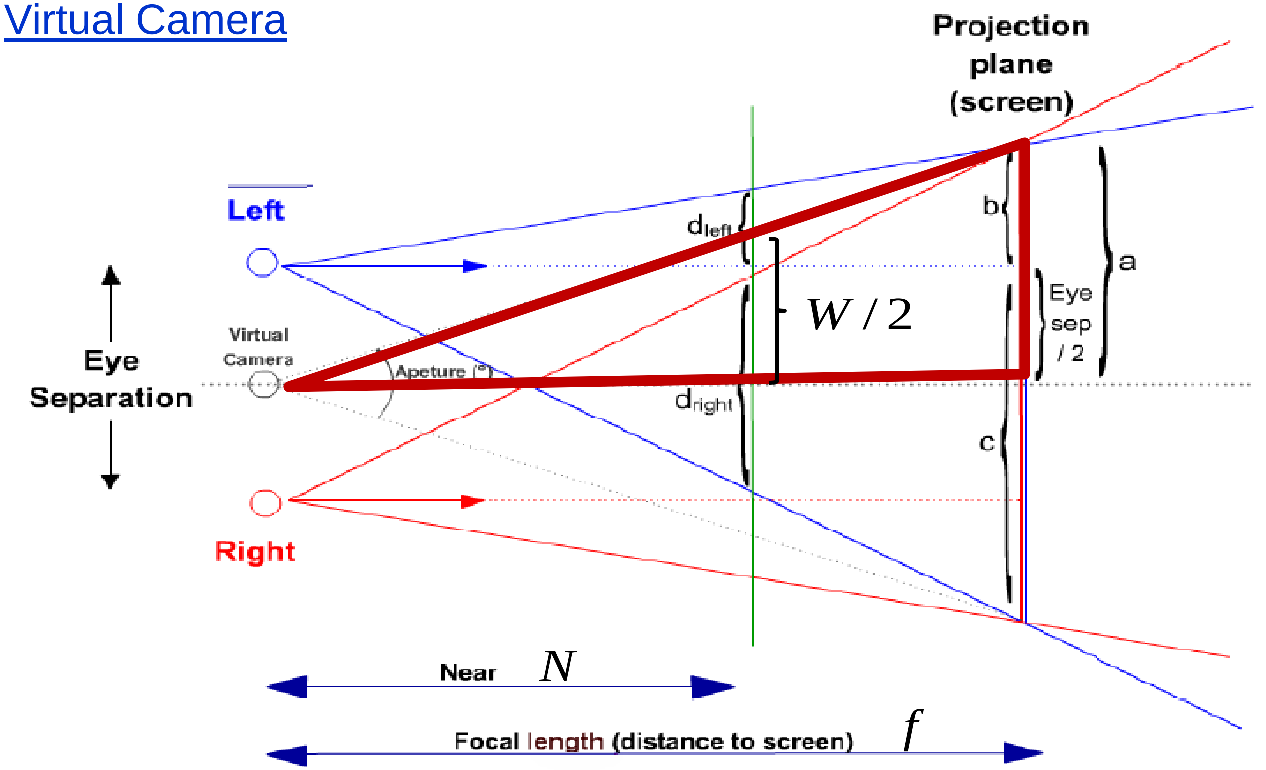
$$\frac{d_{left}}{b} = \frac{N}{f}$$

$$L = -d_{left} = -b\frac{N}{f}$$

$$\frac{d_{right}}{c} = \frac{N}{f}$$

$$R = d_{right} = c\frac{N}{f}$$

Virtual Camera



$$\frac{d_{left}}{b} = \frac{N}{f}$$

$$\frac{d_{right}}{c} = \frac{N}{f}$$

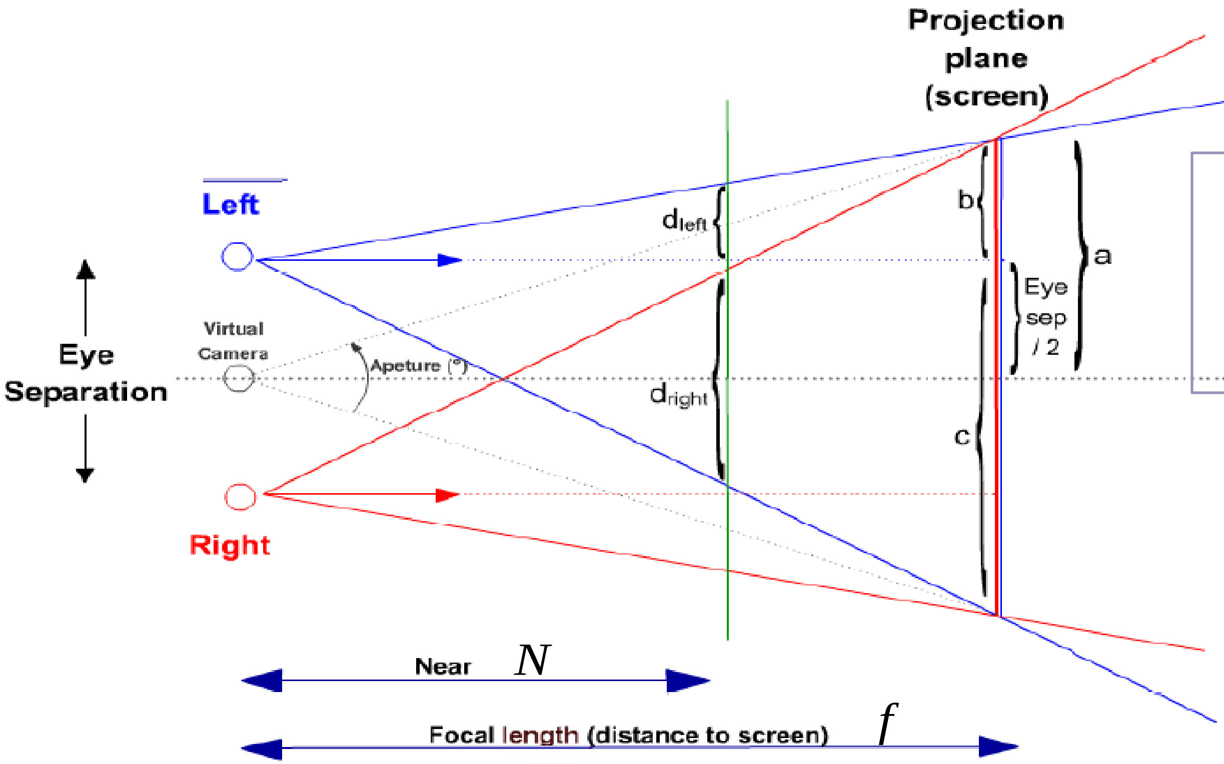
$$\frac{a}{f} = \frac{W/2}{N} = \tan(\theta/2)$$

$$a = f.\tan(\theta/2)$$

$$a = f.\frac{W/2}{N}$$

$$L = -d_{left} = -b\frac{N}{f}$$

$$R = d_{right} = c\frac{N}{f}$$



$$b = a - es/2$$

$$c = a + es/2$$

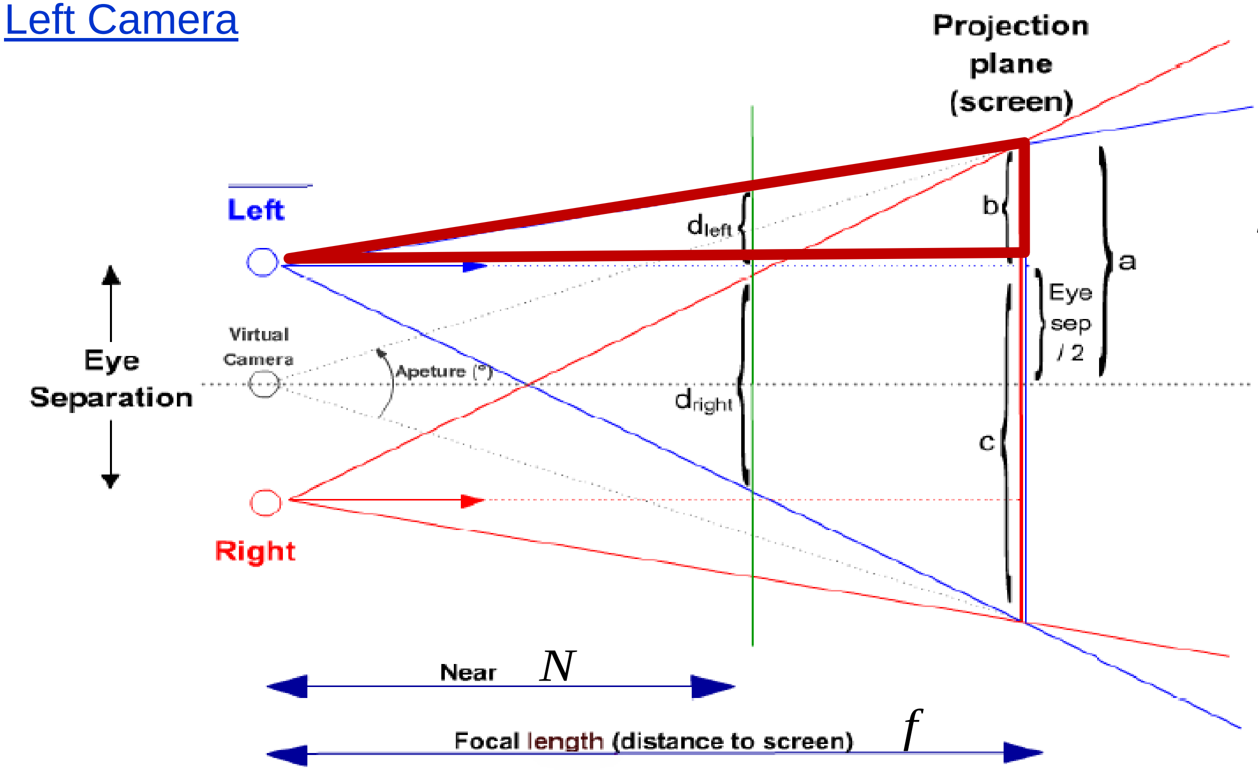
$$\frac{d_{left}}{b} = \frac{N}{f} \qquad \frac{d_{right}}{c} = \frac{N}{f} \qquad \frac{a}{f} = \frac{W/2}{N} = \tan(\theta/2)$$

$$a = f.\tan(\theta/2)$$

$$L = -d_{left} = -b\frac{N}{f} \qquad R = d_{right} = c\frac{N}{f}$$

$$a = f.\frac{W/2}{N}$$

Left Camera



$$b = a - es/2$$

$$c = a + es/2$$

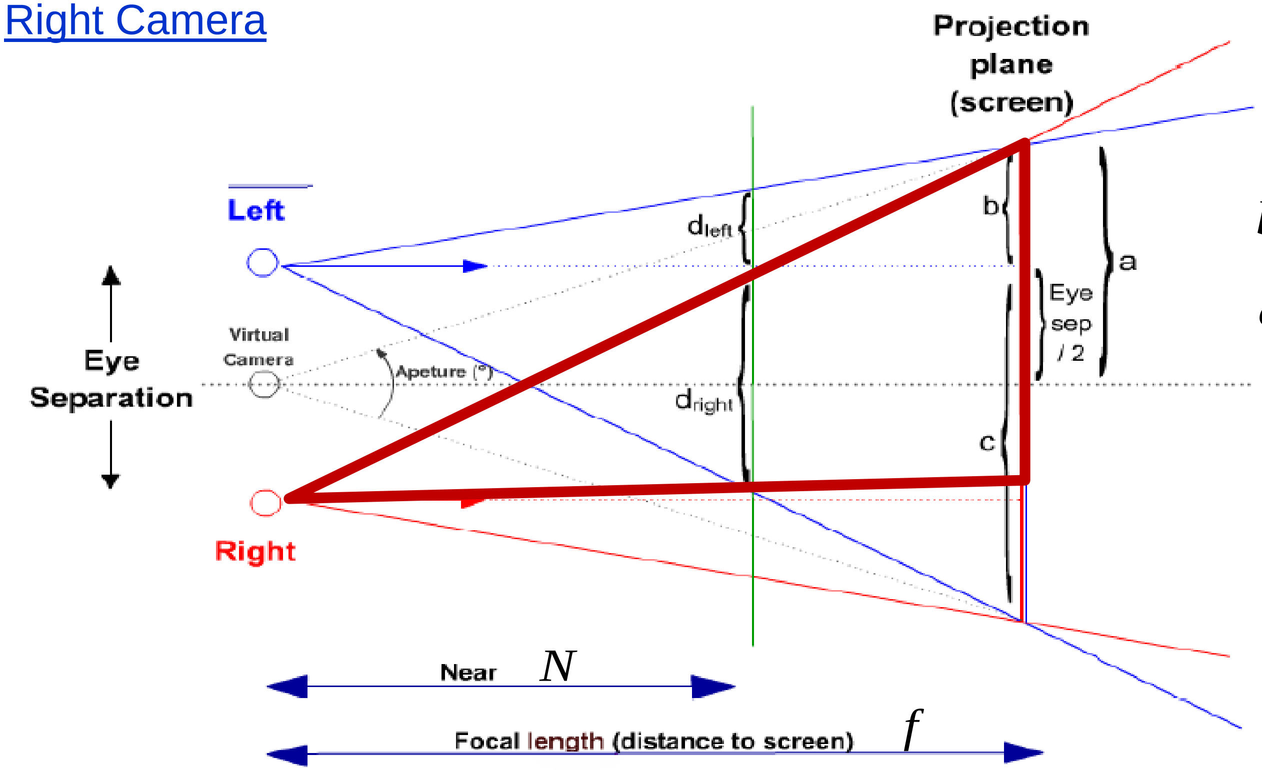
$$a = f.\tan(\theta/2)$$

$$a = f.\frac{W/2}{N}$$

$$L = -d_{left} = -b\frac{N}{f} = -(a - es/2)\frac{N}{f} = -\frac{W}{2} + \frac{es.N}{2f}$$

$$R = d_{right} = c\frac{N}{f} = (a + es/2)\frac{N}{f} = \frac{W}{2} + \frac{es.N}{2f}$$

Right Camera



$$b = a - es/2$$

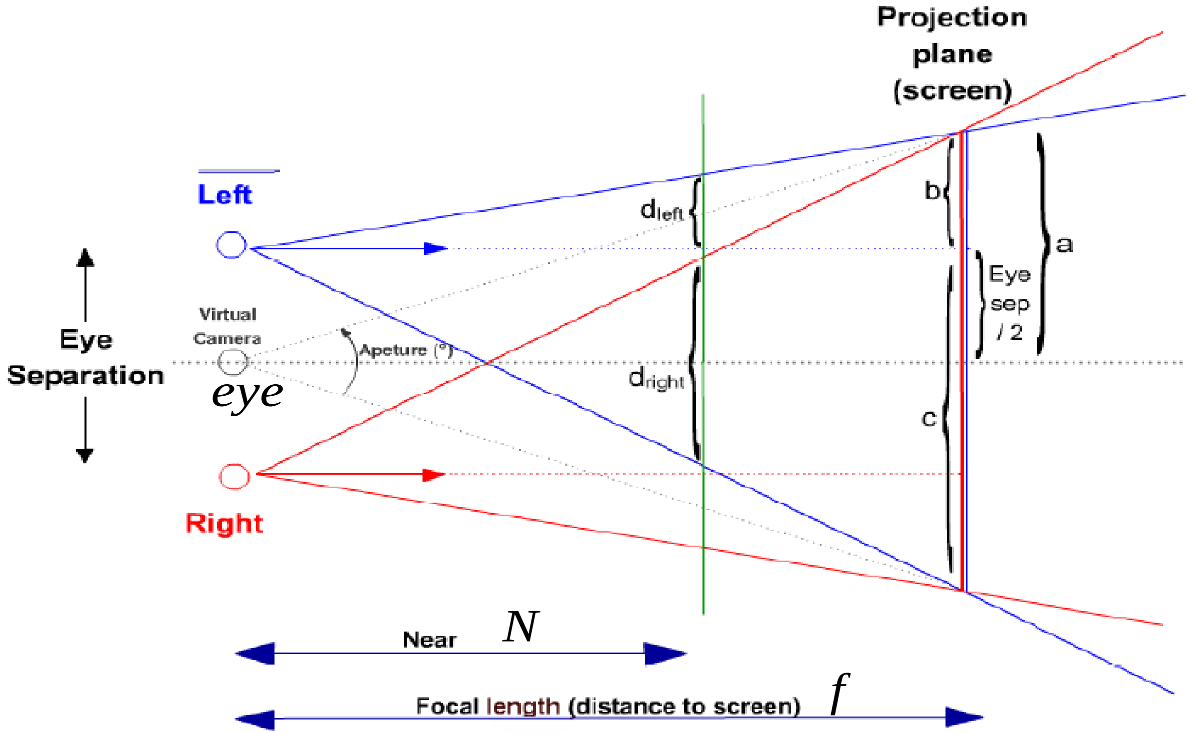
$$c = a + es/2$$

$$a = f.\tan(\theta/2)$$

$$a = f.\frac{W/2}{N}$$

$$L = -d_{right} = -c\frac{N}{f} = -(a+es/2)\frac{N}{f} = -\frac{W}{2} - \frac{es.N}{2f}$$

$$R = d_{left} = b\frac{N}{f} = (a-es/2)\frac{N}{f} = \frac{W}{2} - \frac{es.N}{2f}$$



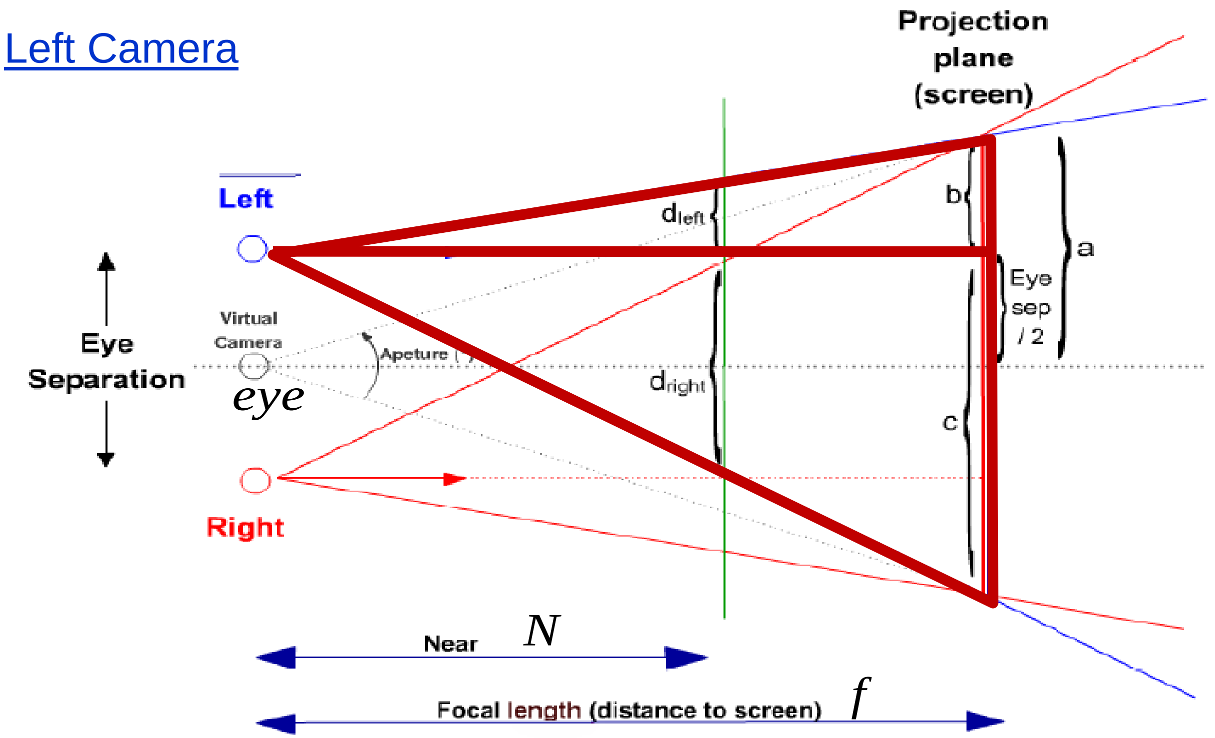
Initially set double buffer

```
glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB | GLUT_DEPTH);
```

In display function, first clear background as:

```
glDrawBuffer(GL_BACK_LEFT);
glClear(GL_COLOR_BUFFER_BIT | GL_DEPTH_BUFFER_BIT);
glDrawBuffer(GL_BACK_RIGHT);
glClear(GL_COLOR_BUFFER_BIT | GL_DEPTH_BUFFER_BIT);
```

Left Camera



$$L = -\frac{W}{2} + \frac{es.N}{2f}$$

$$R = \frac{W}{2} + \frac{es.N}{2f}$$

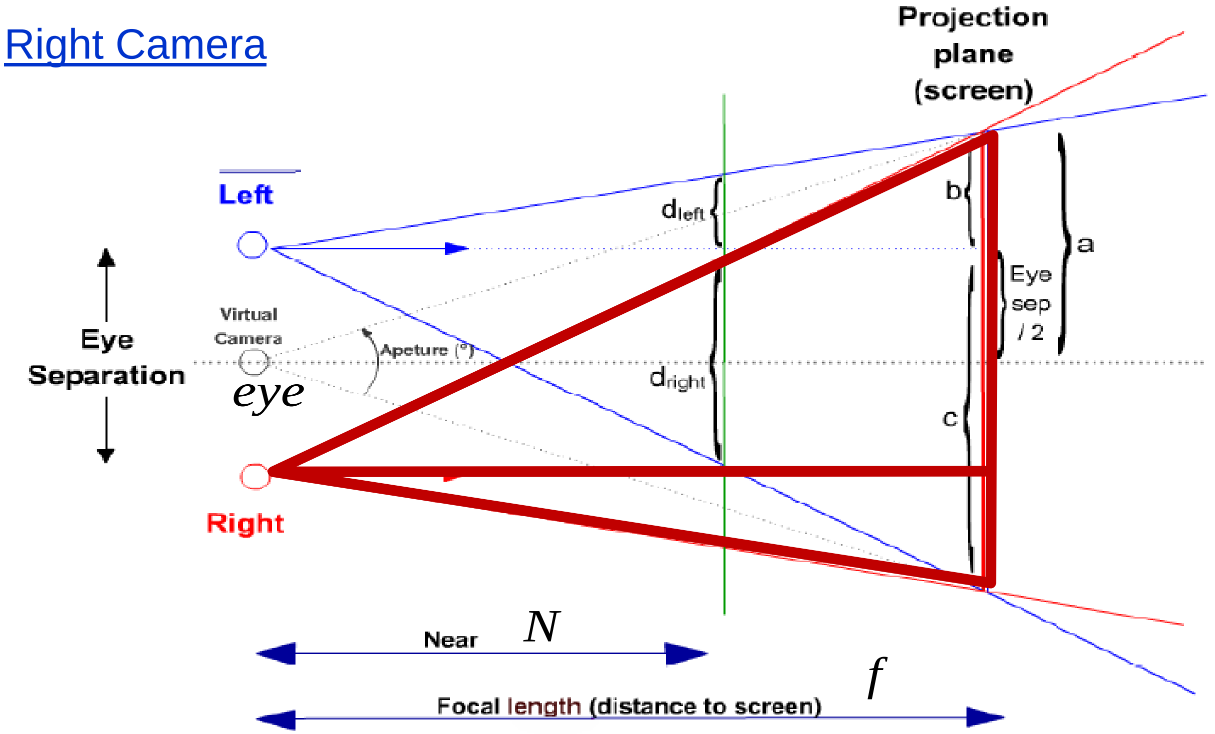
In display function

```
glMatrixMode(GL_PROJECTION); // change to projection matrix
glLoadIdentity();
// set frustum for left eye
L = -W/2 + (N/f) * (es/2.0);  // L for left eye
R =  W/2 + (N/f) * (es/2.0);  // R for left eye
glFrustum(L, R, top, bottom, N, F);

glMatrixMode(GL_MODELVIEW); // change to modelview matrix
glDrawBuffer(GL_BACK_LEFT);
glLoadIdentity();  // locate left camera on the left of the virtual camera
gluLookAt(eye.x - es/2.0, eye.y, eye.z,   eye.x-es/2.0, eye.y, eye.z-10,   0, 1, 0);
glViewport(0, 0, 400, 400);  //set left viewport
apply_light_and_draw_objects();
```

glFrustum (xwmin, xwmax, ywmin, ywmax, dnear, dfar)

Right Camera



$$L = -\frac{W}{2} - \frac{es.N}{2f}$$

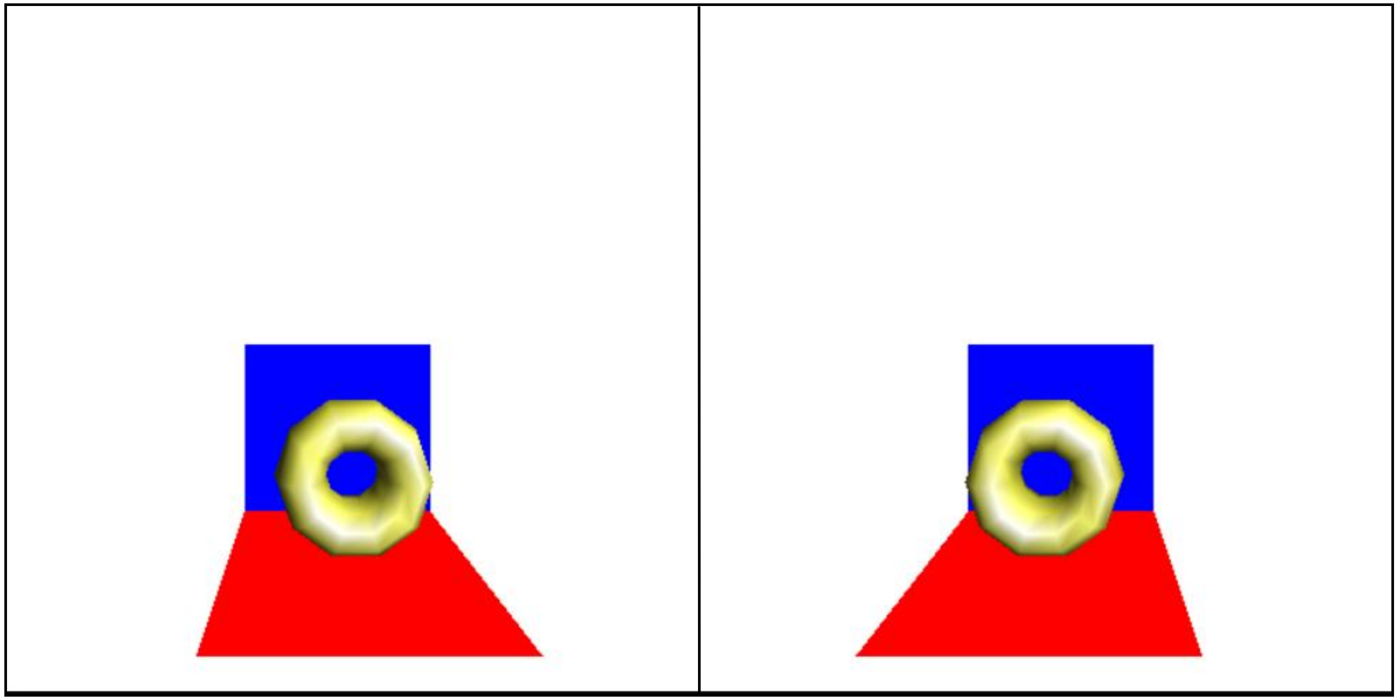
$$R = \frac{W}{2} - \frac{es.N}{2f}$$

In display function

```
glMatrixMode(GL_PROJECTION); // change to projection matrix
glLoadIdentity();
// set frustum for right eye
L = -W/2 - (N/f) * (es/2.0);  // L for right eye
R =  W/2 - (N/f) * (es/2.0);  // R for right eye
glFrustum(L, R, top, bottom, N, F);

glMatrixMode(GL_MODELVIEW); // change to modelview matrix
glDrawBuffer(GL_BACK_RIGHT);
glLoadIdentity();   // locate right camera on the right of the virtual camera
gluLookAt(eye.x + es/2.0, eye.y, eye.z,  eye.x + es/2.0, eye.y, eye.z-10,  0, 1, 0);
glViewport(400, 0, 400, 400); //set right viewport
apply_light_and_draw_objects();

glutSwapBuffers();  // use this instead of glFlush()
```



Example parameters:

```
eye separation (es): 40
focal length (f): 100
Near plane (N): 75
Far plane (F): 200
Width (W): 200
top: -W/2
bottom: W/2
```